

Final Requirements Worksheet Table

#	Requirement (as written)	Source (role/stakeholders)	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes/Resolution
1	The system shall store enough client info such that, in the case an item is damaged or lost, we may follow-up with the client over email.	Shop Owner	FAIL - Conflict with requirement 25	FAIL	PASS	FAIL (item instead of gear)	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 25 for details.
2	The system shall store client identifying info securely (i.e. guard against cyberattacks).	Shop Owner	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: rewrite as “The system shall satisfy PCI-DSS (Payment Card Industry Data Security Standard)”
3	The system shall validate client info, so clients cannot give fake information.	Shop Owner	PASS	FAIL	FAIL	PASS	PASS	PASS	Resolution: rewrite as “The system shall use OTP (One Time Passwords) to verify client identity”
4	The system shall be easily extendable for future developers	Shop Owner	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: rewrite to “The system shall be easily extendable for future development, such as online shopping”
5	The system shall allow us to check the status of gear being rented.	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
6	The system shall track metrics on rented gear and create easy-to-understand graphs and visuals using these metrics.	Shop Owner	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: Split into 2 new requirements: 1. “The system shall record which customer currently has each rented equipment.” 2. “The system shall provide a report on what gear is out right now, what’s overdue, and which items get rented most”
7	The system shall have easy to read guides on how to start the system, stop the system, and troubleshoot errors in the system.	Shop Owner	PASS	FAIL	PASS	PASS	FAIL	PASS	Resolution: Rewrite as “The system shall have guides targeted at nontechnical users for how to start the system, stop the system, and troubleshoot errors in the system”
8	The system shall allow employees to edit gear info in case employees input the wrong information.	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
9	The system shall allow us to toggle which gear/items need deposits and which don’t. This includes manual toggle, as well as applying filters (e.g. all items that cost more than \$200, all items that are larger than 1m^3).	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	

10	The system shall cost no more than \$10 a month to run .	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
11	The system shall be able to back up the store’s system data including client info, documentation, gear information, and more.	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
12	The system shall only allow store employees to view the system’s stored data.	Shop Owner	PASS	PASS	PASS	PASS	PASS	PASS	
13	The system shall be fast and simple to use.	Rental Clerk	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: split it into several requirements: “The system shall take no longer than half a second to verify my credentials, take no longer than half a second to load each page, and take no longer than half a second to reflect any updates I input as an employee.” “The system shall require no more than 15 minutes of onboarding to be used effectively by any employee”
14	The system shall know who previously just used a piece of gear.	Rental Clerk	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: rewrite as “The system shall track the client who previously rented each piece of gear”
15	The system shall be able to search by the name of the client or gear.	Rental Clerk	PASS	FAIL	FAIL	PASS	PASS	PASS	Resolution: Split requirements more clearly “The system shall be able to search up gear items by name as well as the name of rental clients” “The system shall add an additional short numerical identifier for all customers, in case multiple clients have the exact same name.”
16	The system shall be able to flag gear that has been returned damaged as “out of service”.	Rental Clerk	PASS	PASS	PASS	PASS	PASS	PASS	
17	The system shall utilise an ID and deposit-based system when clients take out high value gear to advise clients to be more careful (and to partly pay for damages for high value gear if there are any).	Rental Clerk	FAIL - Conflict with requirement 22	PASS	PASS	PASS	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 25 for details.
18	The system shall give each store item a UUID.	Rental Clerk	PASS	PASS	PASS	PASS	PASS	PASS	
19	The system shall allow clients to rent out multiple equipment at once and return them in any order.	Rental Clerk	PASS	PASS	PASS	PASS	PASS	PASS	
20	The system shall record an item’s condition when it is checked out and when it is returned.	Rental Clerk	PASS	PASS	PASS	PASS	PASS	PASS	
21	The system shall display to clients whether gear they	Customer	PASS	PASS	PASS	PASS	PASS	PASS	

	want to rent is available in real time.								
22	The system shall allow clients to check gear out within several seconds or less.	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
23	The system shall display to clients what gear they are due to return and when to return them in an easy-to-read and intuitive fashion.	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
24	The system shall not use deposits for gear.	Customer	FAIL - Conflict with requirement s 1 & 17	PASS	PASS	PASS	PASS	PASS	Resolution: Replace with two new requirements: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” See Requirement 25 for details.
25	The system shall not ask more from clients other than name and phone number when renting equipment.	Customer	FAIL - Conflict with requirement 1 & 17	PASS	PASS	PASS	PASS	PASS	Resolution: based on client needs of: - Quick checkout of items w/o need for account - Avoiding need to give personal info (as giving too much feels like client has been accused) and store needs of: - Accountability on lost/damaged items - Tracking metrics on who is renting what gear we reached the following compromise: “The system shall identify customers using their name and phone number only, without the need to create an account or password.” “The system shall demand a deposit from customers when checking out certain high-value items.” This ensures clients can check out gear w/o feeling accused, while also satisfying accountability requirements and need for metrics for the store employees. We replace requirements 1, 17, 24, and 25 with these two requirements.
26	The system shall allow clients to easily extend my rentals.	Customer	PASS	FAIL	FAIL	PASS	FAIL	PASS	Resolution: rewrite as “The system shall allow clients to extend their rentals for an additional 3 days or further for additional charges in a visually clear and responsive manner (if the rented gear is available for rent)
27	The system shall keep clients informed of upcoming due	Customer	PASS	PASS	PASS	PASS	PASS	PASS	

	dates for rented gear.								
28	The system shall be online during store times so that clients may return their gear and have the return be properly logged.	Customer	PASS	PASS	PASS	PASS	PASS	PASS	
29	The system shall support about 150 items and about 50 rentals per weekend day without slowing down.	(AI) Scale and Simplicity	PASS	FAIL	PASS	FAIL	PASS	PASS	Resolution: “without slowing down” is ambiguous, and “items of gear” should be used instead of “items”. Rewrite: The system shall support about 150 pieces of gear and about 50 rentals per weekend day while maintaining at least 97% of its baseline speed when loading and updating rentals and pieces of gear.
30	Each item shall have a unique ID, such as a tag or sticker number, so identical items (e.g., three of the same tent) can be told apart.	(AI) Gear identification and condition	PASS	PASS	PASS	PASS	PASS	PASS	
31	A single rental shall be able to include multiple items. For example, a family might rent two kayaks, paddles, and life jackets together.	(AI) Customers	PASS	PASS	PASS	PASS	PASS	PASS	
32	The system shall support partial returns, where some items in a rental come back earlier than others.	(AI) Operations	PASS	FAIL	PASS	PASS	PASS	PASS	Resolution: requirement is too vague “The system shall support early item returns for certain singular rentals that include multiple items.”
33	Customer data shall be stored securely and viewable only by staff.	(AI) Non-functional	PASS	PASS	PASS	PASS	PASS	PASS	
34	Data shall be backed up so records survive a device failure.	(AI) Non-functional	PASS	PASS	PASS	PASS	PASS	PASS	